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PROFILE SUMMARY

Master's student in Medical Technology at IIT Jodhpur with strong foundations in software development and data-driven systems. Experienced in building backend applications, processing scientific datasets, and developing real-time signal processing pipelines using Python and JavaScript.

Hands-on experience in developing end-to-end solutions, including REST APIs, data preprocessing workflows, and hardware-integrated systems such as EMG-based controllers. Proficient in Node.js, Express, and modern web technologies, with additional exposure to biomedical data analysis and validation workflows.

Background in software testing and product development enables writing reliable, well-documented code and working effectively in Agile, cross-functional teams. Interested in backend engineering, system design, and data-intensive applications.

TECHNICAL SKILLS

Programming

Java, Python

Backend

Node.js, Express.js

Web

HTML, CSS, JavaScript

Cloud & Tools

GitHub, Postman, AWS (S3, EC2)

Data Analysis

NumPy, Pandas, Matplotlib

Operating Systems:

Linux (Ubuntu, basic command-line operations, process/file management)

IoT & Embedded Systems:

Arduino, Raspberry Pi, Sensor Integration, Serial Communication

Projects

SERS-Based Diagnostic Platform (Raman Spectroscopy Analysis)

Tech Stack: Python, NumPy, Pandas, Matplotlib

- Developed data preprocessing pipelines for Raman spectra collected using Ag/Au-coated silicon SERS substrates
- Performed baseline correction (ALS), smoothing (Savitzky-Golay), and noise reduction for improved spectral clarity
- Conducted peak detection and spectral feature extraction for biomarker identification
- Performed Raman spectral analysis of N-Acetylmuramic Acid (NAM) through both simulation and experimental powder measurements
- Compared simulated spectra with experimental data to validate peak positions and functional group assignments
- Identified characteristic peaks and mapped them to corresponding molecular vibrations
- Executed experimental Raman analysis of powder samples and ensured consistency through preprocessing and validation workflows
- Analysed spectral signatures of inflammatory and sepsis biomarkers (CRP, IL-6, TNF- α)
- Contributed to development of label-free diagnostic workflows for early disease detection
- Ensured data accuracy, reproducibility, and consistency across experimental datasets

TourRx - Travel Healthcare Support Platform (tourrx.com)

Tech Stack: HTML, CSS, JavaScript

- Contributed to frontend development for a platform assisting Indian tourists abroad with teleconsultation and emergency support.

- Built mobile-friendly UI components and integrated interactive location-based features.

Kyron Healthcare Website Management

Tech Stack: HTML, CSS, JavaScript

- Managing and maintaining the Kyron Healthcare website, ensuring consistent UI and content updates.
- Improving page structure, asset optimization, and layout flows to enhance user experience.
- Implementing analytics-based adjustments to improve site engagement.

NeuroSync – EMG-Based VR Game Controller

Tech Stack: Arduino Uno, BioAmp EXG Pills, Unity 3D, Signal Processing

- Developed a real-time neuromuscular control system using EMG signals for interaction with a Unity-based VR football game
- Acquired biosignals using multiple EXG sensors placed on wrist, fingers, and lower limb muscles
- Designed a custom signal-processing pipeline including amplification, filtering, envelope detection, and peak confirmation
- Implemented auto-calibration and rise-only triggering to minimize noise and false positives• Translated muscle activation patterns into control commands (movement + kick actions)
- Streamed real-time data from Arduino to Unity for interactive gameplay
- Built a functional prototype enabling controller-free human-computer interaction

INTERNSHIP

10/2024 – 03/2025	Intern Trainee - Product Management <i>Jodhpur City Knowledge and Innovation Foundation (JCKIF)</i> • Managed digital product catalog, ensuring accurate listings and metadata consistency. • Captured and edited product images to enhance visual quality and user appeal. • Supported backend workflows and coordinated with teams to maintain platform performance.
10/2022 – 11/2022	Intern Trainee - Software Testing <i>Atos Syntel</i> • Assisted in manual and automation testing within an Agile development environment. • Performed functional, integration, and regression testing across application modules. • Collaborated with development teams for bug tracking, verification, and documentation.

EDUCATION

2024 – Present Rajasthan, India	M.Tech in Medical Technology <i>Indian Institute of Technology Jodhpur</i> CGPA-8.20/10
2022 – 2024 Tamilnadu, India	Master Of Computer Applications <i>Erode Sengunthar Engineering College</i> CGPA-7.20/10
2019 – 2022 Tamilnadu, India	Information Technology <i>Kongu Arts And Science College</i> CGPA - 6.9/10

ACHIEVEMENTS & AWARDS

Finalist Smart India Hackathon 2024

1st Prize - Website Creation | 2nd Prize - Code Fighter

Bronze - Inter-Collegiate Taekwondo

Represented Tamil Nadu - Senior National Softball Championship